

# CSE107 — Lab #10

Lab Date: 12/12/2025 — Due: next lab day

|                                    |
|------------------------------------|
| <b>Student: Zeynep Özge Toptaş</b> |
| <b>Student ID: 240104004006</b>    |
| <b>Section: Section 2</b>          |
| <b>Instructor: Zeliha Erim</b>     |

## 1. Objectives

This lab practiced using C language functions like `strlen` and implemented sorting algorithms (Bubble Sort and Quick Sort) to analyze student data.

## 2. Describe and Use Functions

### Strlen:

- `strlen` function calculates the length of a string.
- `size_t strlen(const char *str)`
- The function returns the number of characters in the string
- `size_t` is an unsigned integer type used to represent the size of objects in memory.
- The function counts characters starting from the beginning until it finds the null terminator (`\0`).
- This function belongs to the `<string.h>` library.

### Strcmp:

- `strcmp` function compares two strings lexicographically (based on ASCII values).
- `int strcmp(const char *str1, const char *str2)`
- This function returns 0 if strings are equal, a negative value if `str1` is less than `str2`, or a positive value if `str1` is greater than `str2`.
- Two constant character pointers: `str1` (first string) and `str2` (second string).
- This function compares the strings character by character until they differ or a null terminator is reached.
- The function belongs to the `<string.h>` library.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char str[] = "zozge";
```

```
    printf("length: %zu\n", strlen(str));
```

```
    return (0);
```

```
}
```

```
int main() {
```

```
    char str1[] = "zozge";
```

```
    char str2[] = "zozge";
```

```
    int result_eq = strcmp(str1, str2);
```

```
    printf("result: %d\n", result_eq);
```

### 3. Labwork Evidence — Screenshots

```
6) Show longest
7) Show stars
0) Quit
Make Selection: 7

90-100:* * *
80-89:* *
70-79:* * *
60-69:* * *
0-59:* * * * * * * * * * * * * * * * *

===== Student Score Calculator Menu for 30 Student =====
```